

HV P2P v26.08.31.06

Complete GitHub Build, Arduino IDE & Wiring Guide

CTRL EdgeBox-ESP-100 | W1P EdgeBox-ESP-100 | Waveshare ESP32-S3 7-inch CTRL-TS | macOS SRVR

Release workflow: Upload this revision to the existing **HV-P2P** GitHub repository. The single root workflow builds the firmware and the matching macOS Intel SRVR app. Only after both jobs pass does GitHub create the combined Complete Release artifact.

GitHub output artifacts

Artifact	Contents / use
HV-P2P-v26.08.31.06-Native-Firmware	CTRL-TS, CTRL and W1P compiled binaries plus STAGED_SOURCE. Use STAGED_SOURCE
HV-P2P-SRVR-v26.08.31.06-macOS-Intel	HV P2P SRVR v26.08.31.06 macOS Intel.zip containing HV P2P SRVR.app.
HV-P2P-v26.08.31.06-Complete-Release	Combined matched release containing both the firmware output and the SRVR app ZIP.

The checked-in CTRL source intentionally contains a build guard. GitHub compiles CTRL-TS first, calculates its version/SHA-256, embeds that exact binary into a staged CTRL copy, then compiles CTRL. Therefore the **STAGED_SOURCE** CTRL folder in the GitHub output is the Arduino-IDE-ready CTRL source.

Arduino IDE settings

CTRL and W1P - Seeed EdgeBox-ESP-100

Setting	CTRL / W1P EdgeBox
Board	Edgebox-ESP-100
ESP32 core	3.3.8
CPU Frequency	240 MHz
Flash Mode	QIO 80 MHz
Flash Size	16 MB (128 Mb)
Partition Scheme	Use the sketch-local partitions.csv (dual OTA)
PSRAM	Disabled
USB CDC On Boot	Default / Disabled
USB Mode	Default
Upload Mode	Default / UART0
Upload Speed	921600
Core Debug	None for release; Info may be used temporarily
Erase All Flash	Enabled for first commissioning flash; then Disabled

CTRL-TS - Waveshare ESP32-S3 7-inch

Setting	Waveshare CTRL-TS
Board	ESP32S3 Dev Module (matches native GitHub build)
ESP32 core	3.3.8
CPU Frequency	240 MHz
Flash Mode	QIO 80 MHz
Flash Size	16 MB (128 Mb)
Partition Scheme	Use the sketch-local partitions.csv (dual OTA)
PSRAM	OPI PSRAM
USB CDC On Boot	Enabled
USB Mode	Hardware CDC and JTAG
Upload Mode	Default / UART0 / Hardware CDC
Upload Speed	921600
Core Debug	None for release; Info may be used temporarily
Erase All Flash	Enabled for first commissioning flash; then Disabled

Keep every file in each STAGED_SOURCE sketch folder together, especially partitions.csv, the RS485 header, and the generated CTRL-TS firmware-image header inside the staged CTRL folder.

Low-voltage field wiring

Node / function	Controller connection	External connection
CTRL joystick signal	AI0	0-5 V Hall joystick output
CTRL joystick reference	AGND	Joystick signal 0 V
CTRL joystick supply	External regulated 5 V	Do not apply 24 V to joystick
CTRL local E-stop status	DI0	24 V when healthy; open / 0 V when pressed/broken loop
CTRL DI common	S/S	Recommended: S/S to 0 V; +24 V -> NC E-stop -> DI0
CTRL -> CTRL-TS	RS485 A / B	A -> Waveshare A, B -> Waveshare B
W1P local E-stop status	DI0	24 V when healthy; open / 0 V when pressed/broken loop
W1P DI common	S/S	Recommended: S/S to 0 V; +24 V -> NC E-stop -> DI0
W1P -> Leadshine EL7	RS485 A / B	A -> EL7 485+, B -> EL7 485-
CTRL/W1P Ethernet	RJ45	HV P2P LAN / SRVR

Safety: DI0 is a supervisory software E-stop status input. It is not a substitute for a hardwired machine E-stop / STO / contactor / brake safety chain. Keep powered motion disabled until communications, limits, E-stop states and drive feedback have all been verified.

Network defaults

SRVR	172.20.1.100
CTRL	172.20.1.101
W1P	172.20.1.102
Subnet	255.255.0.0
Service port	5000

First commissioning order

1. Run the complete GitHub workflow and require all three jobs to pass.
2. Download HV-P2P-v26.08.31.06-Complete-Release (or the two component artifacts).
3. Use FIRMWARE/STAGED_SOURCE to flash CTRL-TS first in Arduino IDE.
4. Flash CTRL EdgeBox, then connect CTRL <-> CTRL-TS RS485 and Ethernet with motion disabled.
5. Confirm CTRL-TS reports the same v26.08.31.06 identity and CTRL does not request an unnecessary update.
6. Verify CTRL E-stop and joystick Left/Centre/Right calibration.
7. Flash W1P EdgeBox and connect W1P <-> EL7 RS485 with motor motion disabled.
8. Start the matching HV P2P SRVR.app from the SRVR artifact and verify CTRL/W1P/HMI status.
9. Verify limits, E-stop sources, brake/drive state and position feedback before any powered motion.

Future revision workflow

For future changes, the supplied revision ZIP is what you upload to the existing HV-P2P GitHub repository. A successful root Actions workflow produces the firmware, the matching SRVR Mac app and one combined Complete Release artifact. Arduino IDE remains the normal tool used to flash the physical ESP32 devices.